

Thermal modeling of cylindrical lithium ion battery during discharge cycle

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ABSTRACT

Transient and thermo-electric finite element analysis (FEA) of cylindrical lithium ion (Li-ion) battery was presented. The simplified model by adopting a cylindrical coordinate was employed. This model provides the thermal behavior of Li-ion battery during discharge cycle. The mathematical model solves conservation of energy considering heat generations due to both joule heating and entropy change. A LiCoO₂/C battery at various discharge rates was investigated. The temperature profile from simulation had similar tendency with experiment. The temperature profile was decomposed with contributions of each heat sources and was presented at several discharge rates. It was found that the contribution of heat source due to joule heating was significant at a high discharge rate, whereas that due to entropy change was dominant at a low discharge rate. Also the effect of cooling condition and the LiNiCoMnO₂/C battery were analyzed for the purpose of temperature reduction.

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1. Introduction

The development of secondary batteries is presently focusing on the rechargeable lithium-ion (Li-ion) cells for their high voltage and low self-discharge rate [1,2]. The Li-ion batteries attract much interest as an energy storage device since they are capable of providing high energy density and wide range of its application. However high fabrication cost, long-term stability and poor safety characteristics caused by high heat generation during charge/discharge cycles are the challenges [2]. Furthermore excessive temperature rise at a high discharge rate will induce safety issue which may lead to performance degradation and thermal runaway, even the cell burning. In order to scale-up for the automotive application, thermal stability issues must be overcome. An achieving of good battery thermal management would encourage the commercialization of hybrid electric vehicle (HEV) and electric vehicle (EV).

Several recent researches have been studied to understand the thermal behavior of Li-ion batteries by experimental investigation along with numerical simulation. Bernardi et al. [3] described the general energy balance for battery systems by assuming uniform temperature throughout the cell. Doyle et al. [4] developed micro-scale model for the behavior of lithium/polymer/insertion cell under the isotropic condition. Chen and Evans [5], Newman and Tiedemann [6], Pals and Newman [7], Doyle et al. [8], and Thomas and Newman [9] were presented mathematical model focusing on

the thermal management. Hong et al. [10] indicated that the entropy change can contribute more than 50% of the total heat generated at the 1 C discharge rate. Srinivasan and Wang [11] studied electrochemical and thermal behavior of Li-ion cells employing two-dimensional unit layer. They found that the effect of heat generation due to entropy is important at all C-rates, especially at low C-rate. The determination of entropy change has been carried out by several researchers. The entropy change of LiCoO₂ and LiC₆ was measured by Reynier et al. [12,13], and that of LiMn₂O₄ was determined by Kim and Pyun [14], and Yazami et al. [15]. Measuring of thermodynamic properties and entropy changes which are recently available is significantly contributed to the development of thermal management of Li-ion battery. Williford et al. [16] and Viswanathan et al. [17] investigated the effect of entropy changes on the thermal behavior of Li-ion batteries. They pointed out that entropy change of LiCoO₂ is larger than the other cathode materials. Pesaran [18] developed lumped capacitance battery thermal model in Matlab/Simulink environment. However he did not describe in details for the mathematical technique. Kim et al. [19] developed thermal abuse model of Li-ion battery by comparing a simple lumped model. Kim et al. [20,21] investigated thermal behavior of Li-ion polymer battery (LIPB). They predicted temperature distribution as a function of potential and current density for the scale-up of the LIPB. However they did not describe how the entropy changes are taken into account on their simulation. Onda et al. [22,23] investigated thermal behavior of Li-ion cylindrical battery. They figured out that temperature may increase above allowable limits during rapid charge/discharge cycles. Inui et al. [24] simulated cylindrical and prismatic Li-ion batteries and validated with experimental results. They modeled commercially available SONY-US18650 cylindrical battery. However they did

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Nomenclature

A	area, m^2	T	temperature, K or $^{\circ}\text{C}$
C_p	heat capacity, J/kg K	V	voltage, V
F	Faraday constant, $96,485 \text{ C/mol}$	V°	open circuit voltage, V
ΔG	gibbs energy change, J/kmol	<i>Greek letters</i>	
h	heat transfer coefficient, W/m K	κ	thermal conductivity, W/m K
i	current density, A/m^2	ρ	density, kg/m^3
n	the charge number pertaining to the reaction	σ	electrical conductivity, S/m
$\dot{q}$	heat generation		
ΔS	entropy change		

not consider the effect of each component in the jelly-roll, instead they simplified those as a lumped region. So far, there are a great deal of current research on secondary batteries, especially material characteristics and structures. However relatively little of the literature concentrates on the thermal simulation with commercial Li-ion batteries.

In this study, the thermal behavior of cylindrical Li-ion battery is numerically investigated for the purpose of battery thermal management. The transient and thermo-electric model is incorporated with finite element method (FEM) simulation. This paper provides comprehensive understanding about the thermal behavior of Li-ion battery by illustrating temperature distributions and profiles at various discharge rates. The effects of cooling condition and the LiNiCoMnO_2 cathode on the temperature distribution are presented to suggest a temperature reduction.

2. Model description

2.1. Geometry and operating conditions

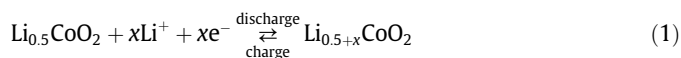
The Li-ion battery considered in this study is illustrated in Fig. 1. A cylindrical SONY-18650 with 1.5 A h capacity is modeled. The basic unit is composed of spirally-wound jelly-roll and can. A jelly-roll is embedded within the cylindrical can and is consist of repeating units of five layers: (1) anode, (2) anode current collector

(copper), (3) separator, (4) cathode, (5) cathode current collector (aluminum). The other components, i.e., tabs, PTC, insulators, etc., are not considered due to minor effect on the thermal analysis of small-size cylindrical battery. Although the jelly-roll is designed with cylindrically spiral type, we modeled it as round structure for the simplicity of the model. The simplified model enables two-dimensional description of the problems by adopting a cylindrical coordinate as shown in Fig. 2. Since the temperature distribution inside the battery is expected symmetric [24], axis-symmetric heat transfer elements are employed for the simulation. A radial boundary conditions at the inside and the convective heat transfer at the outer surface are applied for the boundaries. The geometry details and properties which are collected from open literatures [16,21,25–28] are listed in Table 1.

2.2. Li-ion battery

A typical Li-ion battery uses LiCoO_2/C for cathode/anode materials. The electrochemical reactions during the charge/discharge can be expressed as

At positive electrode:



At negative electrode:

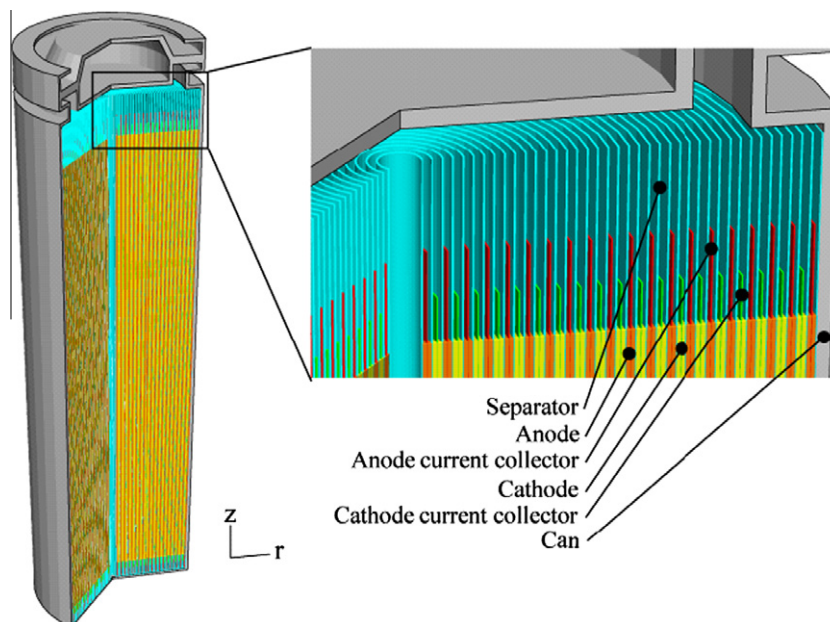


Fig. 1. Schematic illustration of cylindrical Li-ion battery.

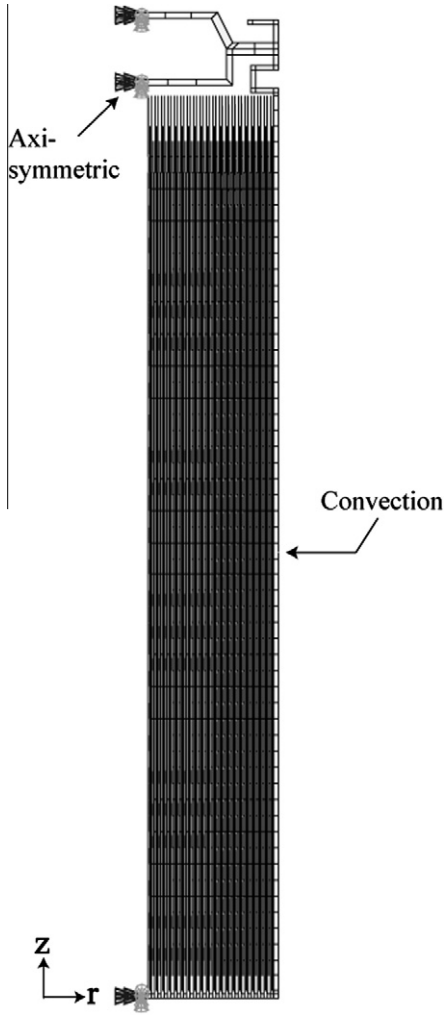
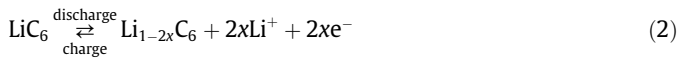
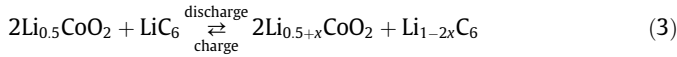


Fig. 2. Numerical model, showing meshes and boundary conditions.



Overall reaction for typical Li-ion battery:



where the reversible reactions proceed from the left to the right during discharge cycle and the opposite direction occurs during charge cycle. $\text{Li}_{0.5}$ is assumed to be 100% state of charge (SOC) of LiCoO_2 by following Williford et al. [16]

During discharge cycle, the heat generations are taken place caused by flowing of current i , through the cell. The heat generation for battery systems which was developed by Bernardi et al. [3] can be written as

$$\dot{q} = i \left(V^o - V - T \frac{\partial V^o}{\partial T} \right) \quad (4)$$

where the term, $i(V^o - V)$, is the heat generation due to joule heating, and the term, $-iT \frac{\partial V^o}{\partial T}$, is the heat generation due to entropy change.

The heat generation due to joule heating can be described as

$$\dot{q}_{\text{joule}} = i(V^o - V) = \frac{i^2}{\sigma} \quad (5)$$

where $(V^o - V)$ is overpotential.

The heat generation due to entropy change can be written as

$$\dot{q}_{\text{entropy}} = -iT \frac{\partial V^o}{\partial T} = -T \Delta S \frac{i}{nF} \quad (6)$$

$$\Delta S = - \frac{\partial \Delta G}{\partial T} = -nF \frac{\partial V^o}{\partial T} \quad (7)$$

where the gibbs energy change is $\Delta G = -nFV^o$.

The entropy changes of LiCoO_2 , LiC_6 and LiNiCoMnO_2 considered in this study are illustrated in Fig. 3. These data are taken from Reynier et al. [12,13] and Williford et al. [16]. It is noticed that the LiCoO_2 , LiC_6 and LiNiCoMnO_2 have negative entropy changes except $\text{SOC} \approx 0.9$ for LiCoO_2 and $\text{SOC} < 0.15$ for LiC_6 . A sixth order least-squares polynomial fit to experimental data has been performed as

$$\Delta S_{\text{Li}_x\text{CoO}_2} = a_0x^6 + a_1x^5 + a_2x^4 + a_3x^3 + a_4x^2 + a_5x + a_6 \quad (8)$$

$$\Delta S_{\text{Li}_x\text{C}_6} = b_0x^6 + b_1x^5 + b_2x^4 + b_3x^3 + b_4x^2 + b_5x + b_6 \quad (9)$$

$$\Delta S_{\text{LiNi}_x\text{Co}_y\text{Mn}_z\text{O}_2} = c_0(\text{SOC})^6 + c_1(\text{SOC})^5 + c_2(\text{SOC})^4 + c_3(\text{SOC})^3 + c_4(\text{SOC})^2 + c_5(\text{SOC}) + c_6 \quad (10)$$

The corresponding coefficients are reported in Table 2. In order to improve approximation, we divided into several sub-regions. The coefficient of determination, R^2 , is used as an indicator of the goodness of approximation. As R^2 indicates, all the empirical equations are close to the data.

By considering the heat generation due to joule heating has a positive value during charge/discharge cycle, the overall discharge reaction can be concluded exothermic.

2.3. Mathematical model

The energy balance equation can be written as

$$\rho C_p A \frac{\partial T}{\partial t} = \nabla(\kappa \nabla T) + \dot{q} \quad (11)$$

$$\dot{q} = \dot{q}_{\text{joule}} + \dot{q}_{\text{entropy}} = \frac{i^2}{\sigma} - T \Delta S \frac{i}{nF} \quad (12)$$

A surface boundary condition which is dependent on the convective heat transfer is defined as

Table 1
Physical properties of materials.

	Thickness (μm)	Height (mm)	Density (kg/m^3)	Thermal conductivity (W/m K)	Specific heat (J/kg K)	Electrical conductivity ($\times 10^6$, S/m)
LiCoO_2	92	53	2291.62	1.85	1.1728	0.0001
LiNiCoMnO_2			1500	5	0.7	0.000139
LiC_6	87	53	5031.67	5	0.7	0.0001
Aluminum	10	55	2700	200	0.87	38
Copper	10	57	9000	380	0.381	60
Separator	22	59	1200	1	0.7	–
Can	300	65	7800	16.8	0.478	–

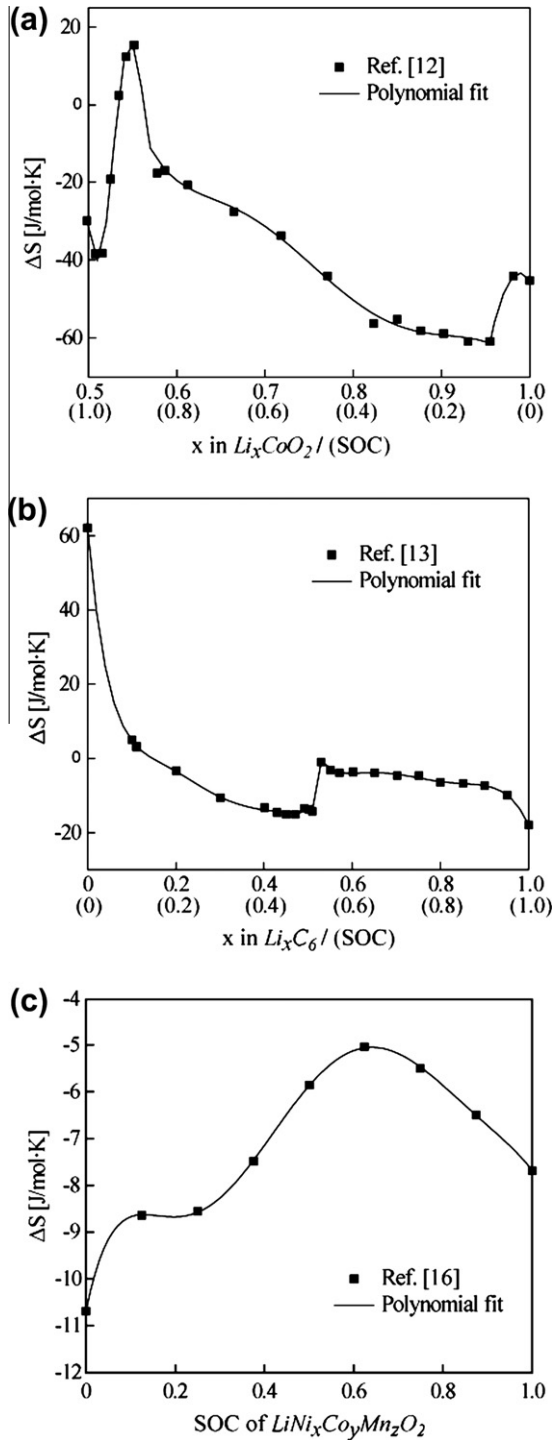


Fig. 3. Entropy changes for (a) LiCoO_2 , (b) LiC_6 , and (c) LiNiCoMnO_2 which are taken from Reynier et al. [12,13] and Williford et al. [16].

$$\dot{q}_{\text{convection}} = h(T - T_o) \quad (13)$$

where h is the heat transfer coefficient. Using a convective heat transfer coefficient permits to incorporate the effect of different cooling rates at different stages of heat treatment.

Boundary conditions for heat transfer analysis in this study using FE modeling are shown in Fig. 2. Convection heat transfer cooling is beginning from room temperature which is applied on outer surface as 300 K for transient thermal analysis. Although potential gradients are expected along the thickness during

Table 2
Polynomial fit to entropy data.

$\Delta S_{\text{Li}_x\text{CoO}_2}$			
x	0.5–0.587	0.587–0.955	0.955–1.0
a_0	–11355177040.8594	150923.273	0
a_1	37082724194.5178	–785628.5568	0
a_2	–50400982654.1226	1674586.0981	0
a_3	36492358554.9118	–1872496.991	0
a_4	–14845027870.6635	1159401.9087	–15259.4369
a_5	–3216993522.9774	–377330.3831	30177.8715
a_6	–290133718.7421	50475.9189	–14963.6876
R^2 ^a	0.9978	0.9957	1.0
$\Delta S_{\text{Li}_x\text{C}_6}$			
x	0–0.509	0.509–0.53	0.53–1
b_0	113503.4266	0	15263.1748
b_1	–213472.7565	0	–90131.2904
b_2	160387.7774	0	207733.4471
b_3	–61018.95	0	244004.4274
b_4	12379.2986	0	155717.1982
b_5	–1340.7464	0.106	–51532.5322
b_6	62.0786	0.4756	6936.5109
R^2 ^a	0.9996	1.0	0.9962
$\Delta S_{\text{LiNi}_x\text{Co}_y\text{Mn}_z\text{O}_2}$			
c_0			–496.66
c_1			1729.4
c_2			–2278
c_3			1382.2
c_4			–380.47
c_5			46.508
c_6			–10.692
R^2 ^a			0.99989

^a Closer to 1 indicates better fits.

discharge, the uniform current generation assumption is considered adequate because the heat generation expression Eq. (4) is developed under the assumption of uniform heat generation throughout the battery system. The left side corresponds to the axis of the axi-symmetric model. The others are considered to be free of any mechanical constraints.

2.4. Numerical method

The transient simulation in the cylindrical battery was performed by ABAQUS finite element analysis (FEA) solver. The above governing equations are discretized by the finite element method (FEM). The four-node structural and quadratic element plane with axi-symmetric option has been used to model the SONY-18650

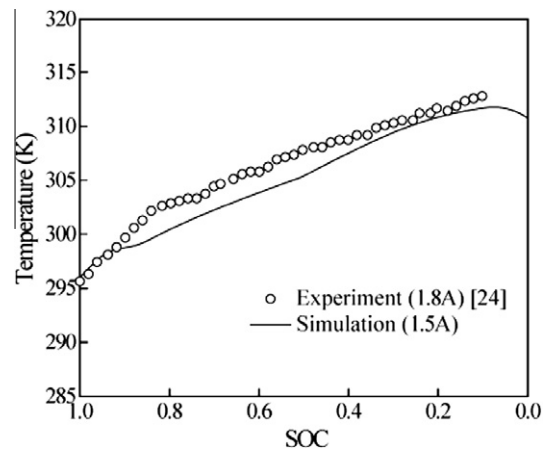


Fig. 4. Comparison between simulation and experiment at a discharge rate of 1 C.

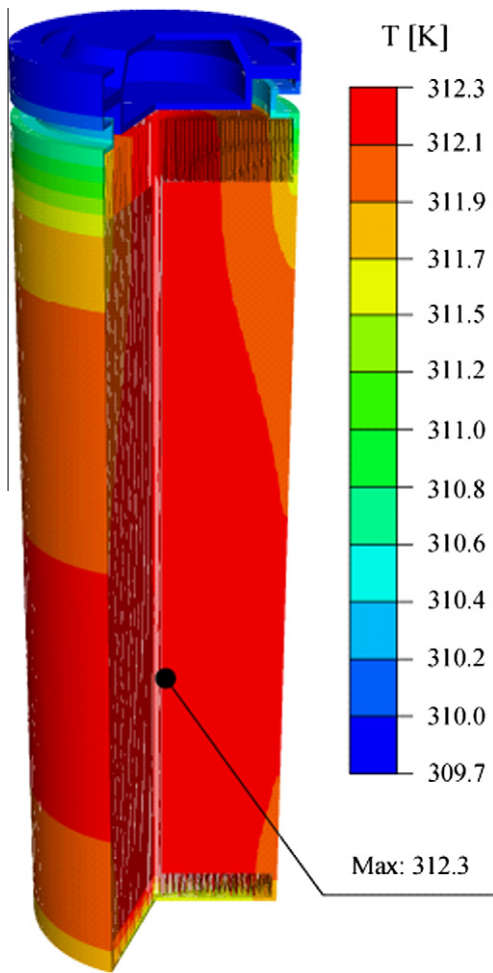


Fig. 5. Temperature distribution of cylindrical Li-ion battery at a discharge rate of 1 C and SOC = 0.1.

battery. The model was meshed with mapped meshing using the quadrilateral-shaped elements. The element size across the plane was uniform since the uniform heat generations are assumed across the area. The fine mesh was imparted at the anodes, cathodes, anode current collectors and cathode current collectors where the heat is generated and it was refined until the results are consistent with only small changes. Time increment satisfying the stability criteria for the constant current discharge is calculated as 1 s. A computer program based on explicit scheme with this time step is developed to compute the temperature field.

3. Results and discussion

The transient FE analysis is performed to investigate the thermal behavior of cylindrical Li-ion battery. SONY-18650 with 1.5 A h capacity that parameters are collected from the literatures is simulated. The battery is operated at constant temperature of 300 K and natural convection heat transfer coefficient of 7.17 W/m K [19]. The heat sources at each components, i.e. anode, cathode, anode current collector, cathode current collector, are taken into account.

3.1. Model validation

The surface temperature of the model is compared with that of the experiment as shown in Fig. 4. Since the data of SONY-18650

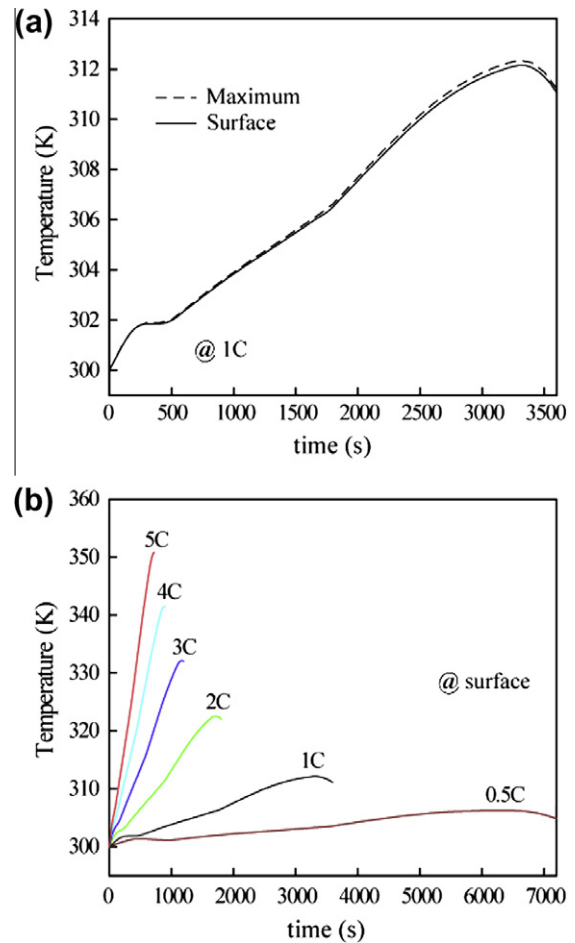


Fig. 6. (a) Comparison of surface temperature and maximum temperature at a discharge rate of 1 C, and (b) surface temperature profiles at discharge rates of 0.5 C, 1 C, 2 C, 3 C, 4 C and 5 C.

with 1.5 A h capacity is not available in the literature, we compare with the experimental result of SONY-18650 with 1.8 A h capacity presented in Inui et al. [24]. The battery is discharged at 1 C rate and the surface temperature is measured at the center of the body. It is observed that the surface temperature of experiment is higher than that of simulation. This is because the battery for experiment has higher capacity than that for simulation. The temperature difference between experiment and simulation is not significant. However the temperature gradient has little discrepancy. This may caused from different material properties and entropy changes according to the changing of capacity. Although the comparison between different capacity batteries is not sufficiently verifiable, we figured out that the simulation result has similar tendency with the experimental data.

3.2. Thermal behavior of Li-ion battery

The temperature distribution of cylindrical Li-ion battery with 1.5 A h capacity is illustrated in Fig. 5. Since the highest surface temperature is presented at SOC = 0.1 as shown in Fig. 4 and the discharge is assumed to be continued until SOC = 0.1 [24] which is correspond to the cut-off voltage, the temperature contour is illustrated at a discharge rate of 1 C and SOC = 0.1. High temperature is observed at the inside of the cell and low temperature is presented at the top cap. The temperature difference between the inside and the outer surface is small. The maximum temperature is found below the center of battery and its rise is 12.3 °C from

Table 3Predicted temperatures in LiCoO₂/C battery at SOC = 0.1.

C-rate	$h = 7.17 \text{ W/m}^2 \text{ K}$			$h = 75 \text{ W/m}^2 \text{ K}$		
	T_{max} (K)	T_{surface} (K)	ΔT	T_{max} (K)	T_{surface} (K)	ΔT
0.5	306.3	306.2	0.1	300.8	300.7	0.1
1	312.3	312.1	0.2	301.6	301.4	0.2
2	322.4	322.1	0.3	304.6	303.9	0.7
3	331.5	331.0	0.5	308.1	307.0	1.1
4	340.3	339.5	0.8	312.3	310.7	1.6
5	348.8	347.8	1.0	317.1	314.9	2.2

the initial temperature. This corresponds well to the result of Chen et al. [29] Moreover the overall appearance of the temperature distribution is good agreement with Inui et al. [24].

Fig. 6 shows temperature profiles at a discharge rate of 1 C. Since it is assumed that the battery is fully charged at the initial state, time = 0 s and 3600 s are corresponds to SOC = 1.0 and 0, respectively. The surface temperature and the maximum temperature are compared in Fig. 6a. The temperature is increased along the discharge time. It is observed that the temperature gradients are changed significantly at time ≈ 330 s, 1800 s and 3300–3600 s, i.e., SOC ≈ 0.9 , 0.5 and 0.1–0. This is because the entropy changes are remarkable at SOC ≈ 0.9 for LiCoO₂ and SOC ≈ 0.5 and 0.1–0 for LiC₆ as shown in Fig. 3a and b. The maximum temperature is slightly higher than the surface temperature. From Table 3, the temperature difference between the maximum and the surface is increased along the discharge rates. However the differences are very small. This is because the heats are generated across the jelly-roll. The surface temperature profiles at discharge rates of 0.5 C, 1 C, 2 C, 3 C, 4 C and 5 C are shown in Fig. 6b. As discharge rate increases, the surface temperature is raised. For low discharge rate, the surface temperature rise is small and the profiles have paraboloidal shape. However for high discharge rate, the surface temperature is increased significantly and the profiles present linear plot. This is due to contributions of each heat sources which will be discussed in the below.

Fig. 7 shows transient temperature contours at a discharge rate of 1 C. The temperature distributions at discharge times of 360 s, 1080 s, 1800 s, 2520 s, 3240 s and 3600 s, i.e., SOC = 0.9, 0.7, 0.5, 0.3, 0.1 and 0, are presented. At the beginning of discharge, the battery has uniform temperature distribution. It becomes non-uniform

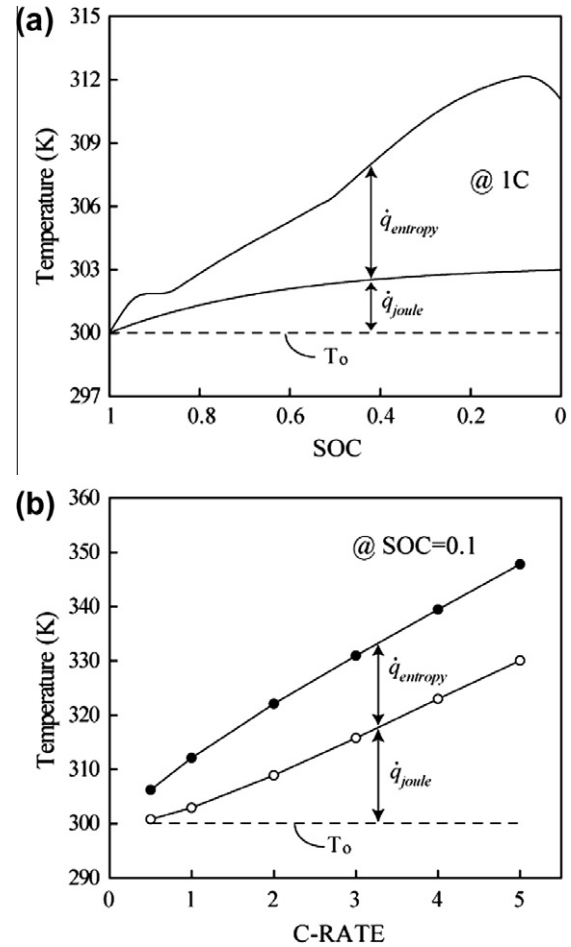


Fig. 8. Surface temperature profile and contribution of each heat sources at (a) a discharge rates of 1 C and (b) SOC = 0.1.

form as increasing of discharge time due to heat dissipation at the surface. However it is noticed that the temperature non-uniformity across the battery is not significant. Since the heat generation occurs at the jelly-roll, high temperature is observed at the inside of the cell.

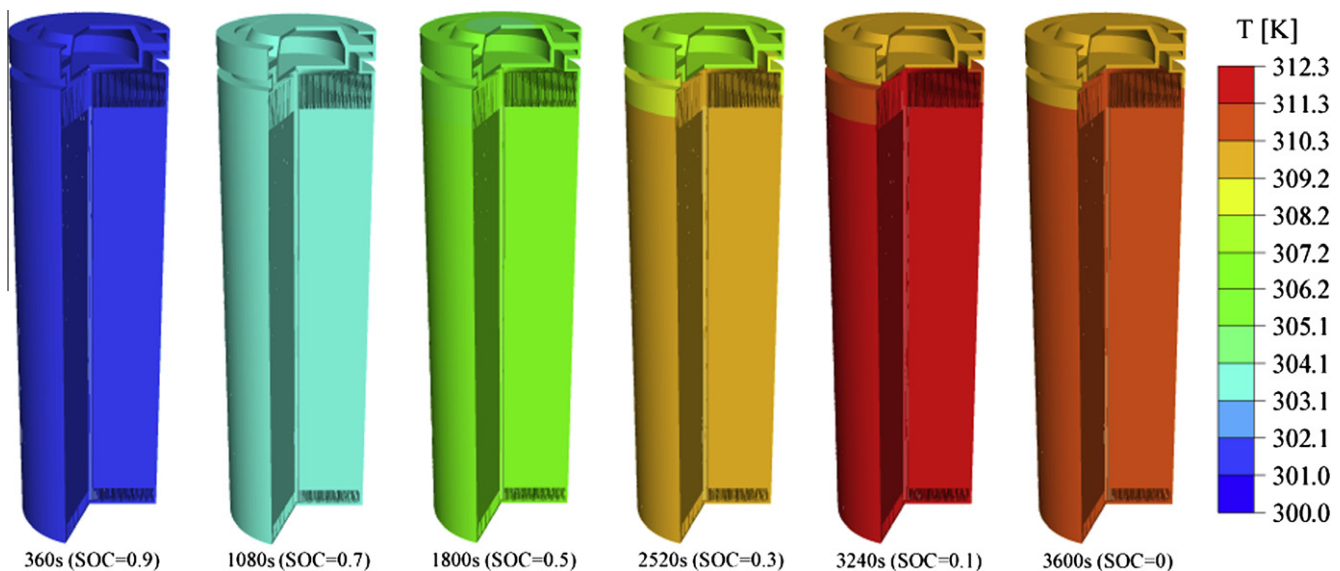


Fig. 7. Simulated sequence of transient temperature contours at a discharge rate of 1 C.

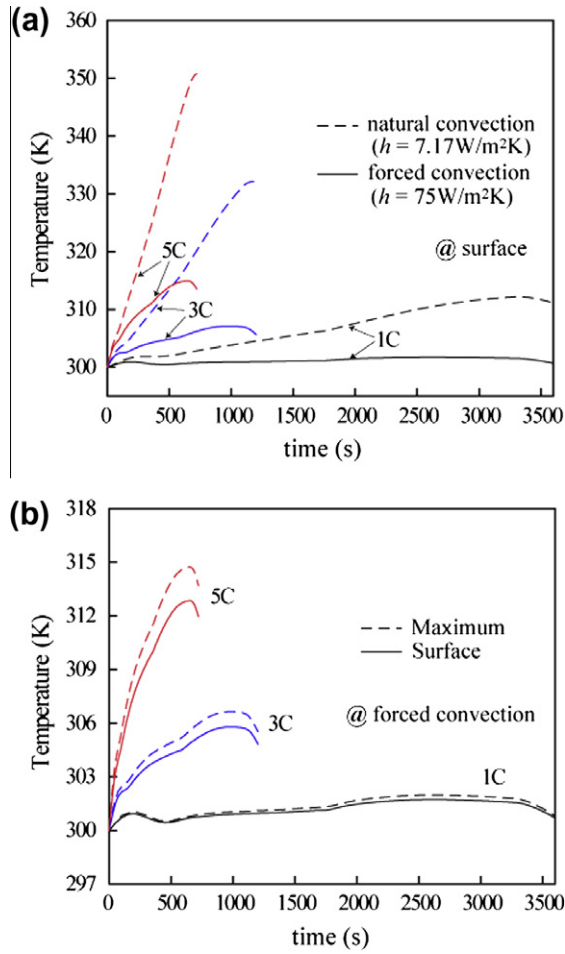


Fig. 9. (a) Surface temperature profiles under different convection conditions and (b) comparison of surface temperature and maximum temperature at discharge rates of 1 C, 3 C and 5 C.

Fig. 8a shows the surface temperature profile and contributions of each heat sources at a discharge rate of 1 C. The contribution of heat generation due to entropy change, $\dot{q}_{\text{entropy}}$, is larger than that due to joule heating, $\dot{q}_{\text{joule}}$, at a discharge rate of 1 C. For high SOC, the difference between the contributions of $\dot{q}_{\text{joule}}$ and $\dot{q}_{\text{entropy}}$ is small. However these differences are increased as decreasing of SOC. At SOC = 0.1, the contributions of $\dot{q}_{\text{joule}}$ and $\dot{q}_{\text{entropy}}$ are estimated 24.1% and 75.9%, respectively. This corresponds well to the result of Hong et al. [10]. It is observed that the contribution of $\dot{q}_{\text{joule}}$ become significant at high discharge rates as shown in Fig. 8b. This figure shows the contributions of each heat sources at SOC = 0.1. At a discharge rate of 0.5 C, the contributions of $\dot{q}_{\text{joule}}$ and $\dot{q}_{\text{entropy}}$ are 12.8% and 87.2%, respectively. However at a discharge rate of 5 C, the contribution of $\dot{q}_{\text{joule}}$ is dominant as its amount is 62.8%, whereas the contribution of $\dot{q}_{\text{entropy}}$ is decreased to 37.2%. This is because the heat sources are strongly dependent on the current density as $\dot{q}_{\text{joule}} \sim i^2$ and $\dot{q}_{\text{entropy}} \sim i$ from Eqs. (5) and (6).

3.3. The effect of cooling condition

From the above, a significant temperature rise is expected at high discharge rates, even risking thermal runaway. Cooling the battery could be effective in depressing the temperature. Fig. 9a shows the surface temperature profiles under different convection conditions at discharge rates of 1 C, 3 C and 5 C. A strongly forced convection, i.e., $h = 75 \text{ W/m}^2\text{K}$ which corresponds to an airflow

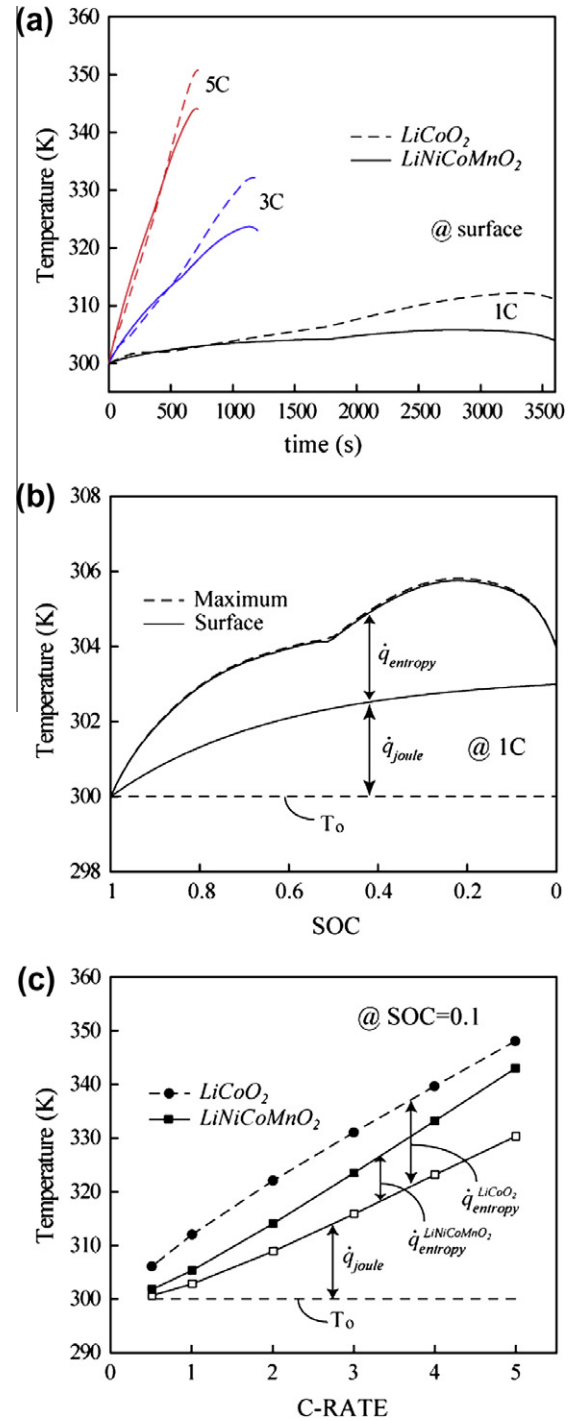


Fig. 10. (a) Surface temperature profiles of LiCoO₂/C battery and LiNiCoMnO₂/C battery at discharge rates of 1 C, 3 C and 5 C, (b) temperature profiles of LiNiCoMnO₂/C battery with contribution of each heat sources at a discharge rates of 1 C, and (c) comparison of LiCoO₂/C battery and LiNiCoMnO₂/C battery with contribution of each heat sources at SOC = 0.1.

of 35 m/s [30], is employed to investigate the effect of cooling condition. After applying a forced convection, the surface temperature is decreased significantly. For SOC = 0.1, the reduced temperature amounts 88.7%, 77.4% and 68.8% at discharge rates of 1 C, 3 C and 5 C, respectively. However the temperature difference between the maximum and the surface is increased as shown in Fig. 9b. From Table 3, the temperature differences are 0.1 °C, 0.2 °C, 0.7 °C, 1.1 °C, 1.6 °C and 2.2 °C at discharge rates of 0.5 C, 1 C, 2 C, 3 C, 4 C and 5 C, respectively. Compared to the results of natural

Table 4
Predicted surface temperature and contribution of each heat sources at SOC = 0.1.

C-rate	LiCoO ₂ /C			LiNiCoMnO ₂ /C		
	T (K)	$\dot{q}_{\text{joule}}$ (%)	$\dot{q}_{\text{entropy}}$ (%)	T (K)	$\dot{q}_{\text{joule}}$ (%)	$\dot{q}_{\text{entropy}}$ (%)
0.5	306.3	12.8	87.2	301.9	41.4	58.6
1	312.3	24.1	75.9	305.4	53.5	46.5
2	322.4	40.4	59.6	314.1	63.3	36.7
3	331.5	51.0	49.0	323.4	67.4	32.6
4	340.3	58.2	41.8	333.0	69.7	30.3
5	348.8	62.8	37.2	342.7	70.3	29.7

convection, the temperature differences between the maximum and the surface are increased high that may result in decreasing of temperature uniformity. Considering that we simulated small-size battery, the thermal management design that minimizes the temperature non-uniformity is required for scale-up to the large-size battery.

3.4. The effect of cathode material

Since the LiCoO₂ has much larger entropy change compared to other cathode materials [17], the layered LiNiCoMnO₂ materials are widely studied for their excellent safety characteristics due to low heat generation [31]. Although the dimensions on the cathode side could be changed by using LiNiCoMnO₂ material, we assumed that the geometry is same as LiCoO₂/C battery. Fig. 10a shows surface temperature profile of two different cathode materials at discharge rates of 1 C, 3 C and 5 C. It is compared that surface temperature of LiCoO₂/C and LiNiCoMnO₂/C batteries. By using LiNiCoMnO₂ material on the cathode, the surface temperature is decreased significantly. The reduced amount is similar at each discharge rates as it is 6.7 °C at 1 C, 7.5 °C at 3 C, and 5.1 °C at 5 C. However for the beginning of discharge, i.e., for high SOC, the amount of temperature reduction by using LiNiCoMnO₂ cathode is very small, even slightly higher temperature than LiCoO₂/C battery at high discharge rates. Fig. 10b shows the temperature profiles of LiNiCoMnO₂/C battery and contribution of each heat sources at a discharge rate of 1 C. The temperatures at the maximum and the surface are compared. As discussed in Fig. 6a, the temperature difference between the maximum and the surface is very small. The temperature gradients are changed significantly at time ≈ 1800 s and 2800–3600 s, i.e., SOC ≈ 0.5 and 0.2–0. These are strongly related to the characteristics of entropy change of both anode and cathode materials as shown in Fig. 3b and c. The contribution of $\dot{q}_{\text{joule}}$ is similar to that of $\dot{q}_{\text{entropy}}$ at a discharge rate of 1 C. For SOC = 0.1, the contributions of $\dot{q}_{\text{joule}}$ and $\dot{q}_{\text{entropy}}$ are estimated 53.5% and 46.5%, respectively. Compared to the LiCoO₂/C battery in Fig. 8a, the contribution of $\dot{q}_{\text{entropy}}$ is decreased much due to low entropy changes of LiNiCoMnO₂ cathode. Fig. 10c shows the comparison of contributions of each heat sources at SOC = 0.1. It is compared that contribution of $\dot{q}_{\text{entropy}}$ in LiCoO₂/C and LiNiCoMnO₂/C batteries. The temperature of LiNiCoMnO₂/C battery is lower than that of LiCoO₂/C battery along the discharge rates. This is due to lower contribution of $\dot{q}_{\text{entropy}}$ in LiNiCoMnO₂/C battery than that in LiCoO₂/C battery due to lower entropy change of LiNiCoMnO₂ cathode. The different temperature gradients between the two batteries are caused by different material properties. From Table 4, for low discharge rate, it is shown that the contribution of $\dot{q}_{\text{entropy}}$ is dominant in LiCoO₂/C battery, whereas the contribution of $\dot{q}_{\text{entropy}}$ is similar to that of $\dot{q}_{\text{joule}}$ in LiNiCoMnO₂/C battery. However for high discharge rate, the contribution of $\dot{q}_{\text{joule}}$ is considerably high, whereas the contribution of $\dot{q}_{\text{entropy}}$ is small. Therefore it can be concluded that the contribution of $\dot{q}_{\text{entropy}}$ both in LiCoO₂/C and LiNiCoMnO₂/C batteries is not significant at high discharge rate. This is because the increasing of heat generation due to joule

heating is higher than that due to entropy change at high discharge rate.

4. Conclusion

The cylindrical Li-ion battery was simulated to provide thermal behavior during discharge cycle. The transient model developed a set of energy equations considering heat generations due to both joule heating and entropy change at each cell components. The comparison with experiment indicated the simulation results has similar tendency with the experimental data. For low discharge rate, the highest temperature was observed at SOC = 0.1 caused by the characteristic of entropy changes of LiCoO₂ and LiC₆. However for high discharge rate, the temperature was increased linearly along the discharge time and the temperature gradient was steep because the contribution of heat generation due to joule heating is high. The contribution of heat generation due to entropy change was dominant at a low discharge rate, whereas that due to joule heating was significant at a high discharge rate. The maximum temperature was observed inside the cell, but the temperature difference between the maximum and the surface was small.

The battery temperature could be reduced by using cooling system and suitable choice of anode and cathode materials which have lower entropy changes. Cooling the battery with forced convection reduced the temperature significantly, but it also increased the temperature non-uniformity that may degrade the battery performance. Using LiNiCoMnO₂ instead of LiCoO₂ on the cathode decreased the temperature, but it was effective at a low discharge rate because the contribution of heat generation due to joule heating is significant at a high discharge rate. Although the current collecting tabs were not considered in this model, it could affect on the results in large-size prismatic battery [16] and LIPB [20,21]. In order to scale-up Li-ion cells to large-size battery, thermal stability problems and non-uniform temperature distribution caused by battery cooling should be overcome.

This study described the thermal behavior of Li-ion battery and the methodology is believed to contribute to the battery thermal design.

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